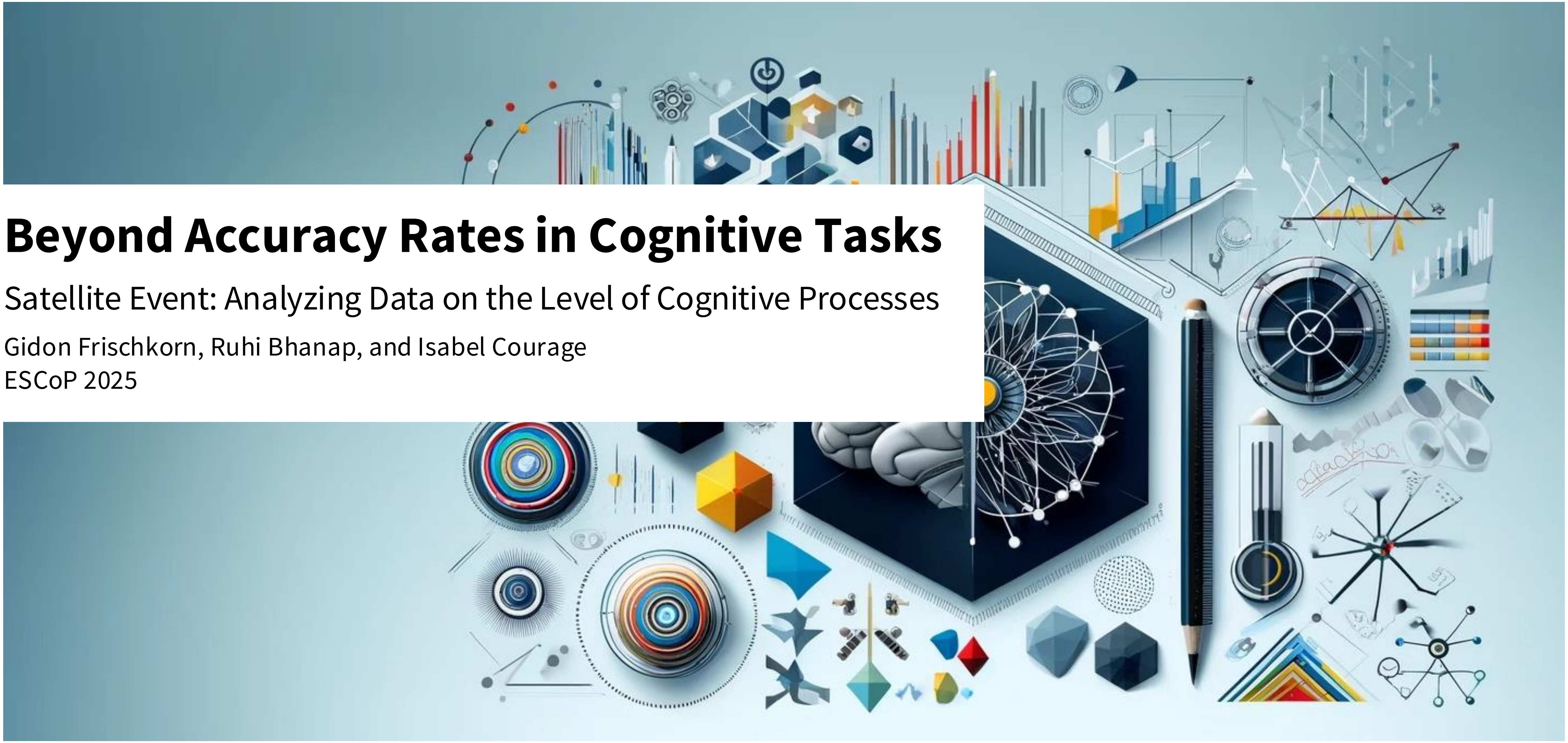
Universität Zürich[UZH]

Department of Psychology



Beyond Accuracy Rates in Cognitive Tasks

Satellite Event: Analyzing Data on the Level of Cognitive Processes

Gidon Frischkorn, Ruhi Bhanap, and Isabel Courage

ESCoP 2025

Workshop: Overview

Monday

Morning

Intro

Why Hierarchical Models?

Signal-Detection Models (brms)

Afternoon

Memory Measurement Model (bmm)

Practical Exercises

Tuesday

Morning

Diffusion Decision Model (bmm)

Practical Exercises

Afternoon

Signal Discrimination Model

Reporting Results from Cognitive Modelling Analyses

Materials are available via GitHub

30 Min. coffee breaks 10:30 & 15:00

60 Min. lunch break 12:30

Ask questions whenever something is not clear ☺

Overview

Memory Measurement Model

- Why errors are so informative about the cognitive processes underlying behavior?
- Introduction to the M^3 framework: Basic M^3 model
- How do we implement and assess M^3 models in bmm?

Why are errors so informative about the cognitive processes underlying behavior?

Memory Measurement Model

- Simple Span Task: Memorize the list of items (e.g., words) in the presented order…

house	dog	shoe	sun	grass	table

- Please recall the the items in the presented order…

?	?	?	?	?	?

Why are errors so informative about the cognitive processes underlying behavior?

Memory Measurement Model

- Knowing what errors participants made, tells us more about the underlying cognitive processes causing the errors
- The same accuracy rates can stem from different kinds of errors

house	**dog**	**shoe**	**sun**	**grass**	**table**

- *Example 1: Accuracy of 67%*

house	**dog**	**sun**	**shoe**	**grass**	**table**

→ All recalled items were presented in the list, but the list positions were mis-memorized

- *Example 2: Accuracy of 67%*

house	**dog**	**sandwich**	**sun**	**leaves**	**table**

→ Some of the recalled items were never presented in the list

Introduction to the M^3 framework

Memory
Measurement
Model

- M^3 framework provides a **set of models** to model the cognitive processes underlying **memory tasks with categorical responses**
  → the type of errors are included to infer the cognitive processes

Type of errors → **Cognitive Processes**

- It includes a **basic model** (for the simple span task), an **extended model** for the complex span task and a **model for a memory updating task** assuming certain cognitive processes

- But the sky is your limit…
  - These models can be applied to other tasks
  - The cognitive processes that are assumed in these models can be adapted and/or expanded

(Oberauer & Lewandowsky, 2019)

Introduction to the M³ framework: Basic M³ Model

Memory Measurement Model

- **Basic model** predicts the frequency of **3 response categories** assuming certain cognitive processes:
  - **Item-in-position (IIP):** correct response
  - **Item-other-position (IOP):** error when an item that was presented in the list at another list position was recalled
  - **Not-presented lure (NPL):** error when an item that was not presented in the list at all was recalled

house	dog	shoe	sun	grass	table
house	dog	sun	shoe	grass	table
house	dog	sandwich	sun	leaves	table

(Oberauer & Lewandowsky, 2019)

Introduction to the M³ framework: Basic M³ Model

Memory Measurement Model

- **Basic model** predicts the frequency **of 3 response categories** assuming certain cognitive processes:
  - **Item-in-position (IIP):** correct response
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What is the error type?

house	dog	shoe	sun	grass	table
hair					
		grass			

(Oberauer & Lewandowsky, 2019)

Introduction to the M³ framework: Basic M³ Model

Memory Measurement Model

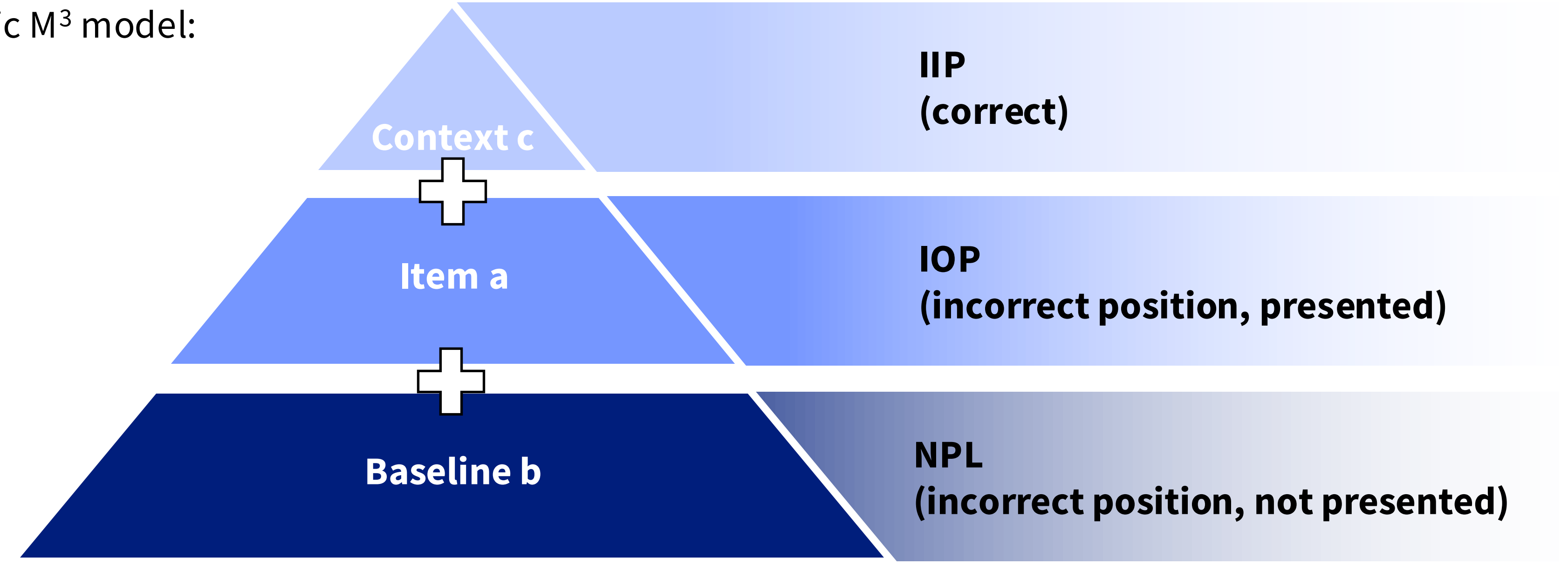
- What **cognitive processes** can be inferred from the response categories in the basic M³ model?
  - **Binding memory / Context activation c** (e.g., position-word binding): estimated parameter
  - **Item memory a** (e.g., word presented in the list): estimated parameter
  - **Baseline activation / Background noise b** (e.g., any word): fixed reference parameter

- Activation functions of basic M³ model:

$$A\,(IIP) \;= b + a + c$$

$$A\,(IOP) \;= b + a$$

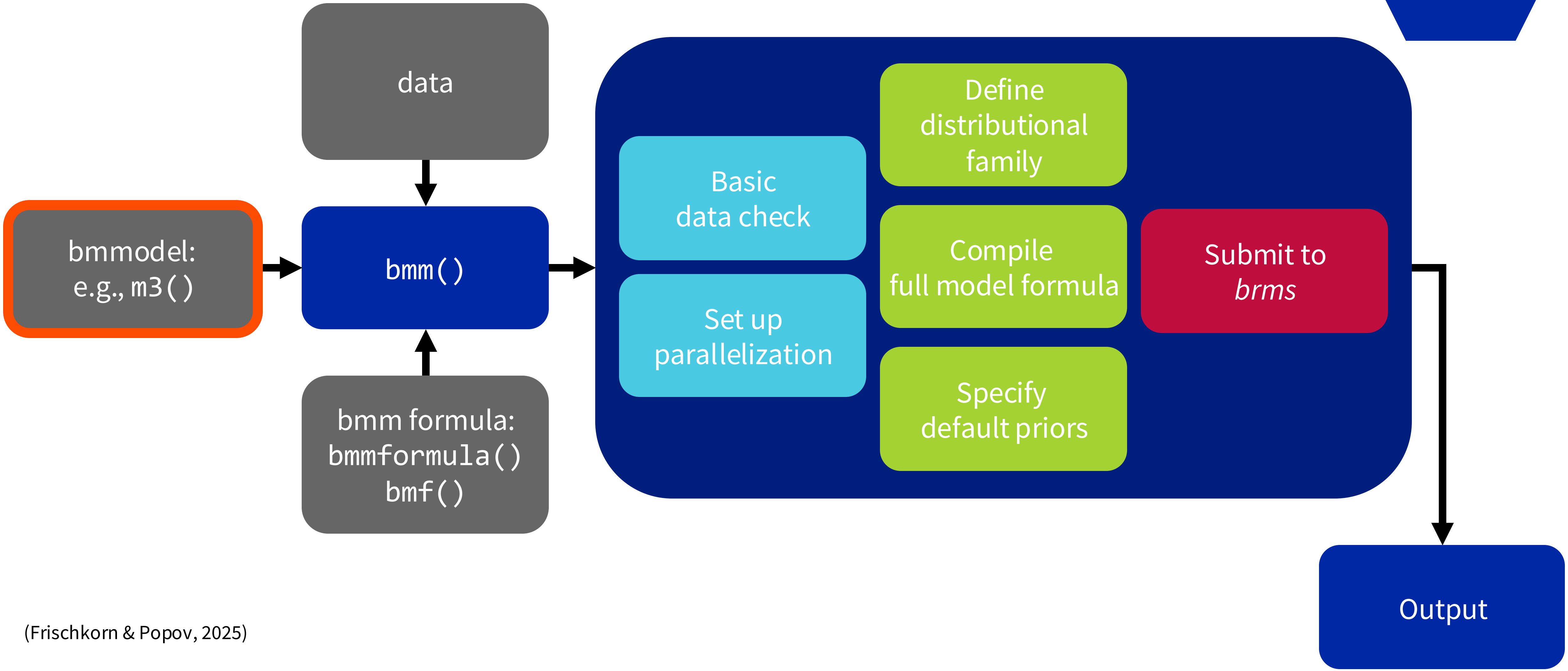
$$A\,(NPL) = b$$



(Oberauer & Lewandowsky, 2019)

How do we implement models in bmm?

Memory Measurement Model



(Frischkorn & Popov, 2025)

How do we implement M^3 models in bmm?
bmmodel

Memory Measurement Model

In *bmm* you set up a bmmodel object for M^3 models using the `m3()` function:

```
m3(
resp_cats,
num_options,
choice_rule,
version
)
```

Response categories: Names of variables that contain the responses for each response category for each trial and/or participant, c("IIP", "IOP", "NPL")
→ can be aggregated over trials

Number of options: Number of options for each response category, c(1, 2, 3)
→ for each single trial

Choice rule: "simple" (simple normalization) or "softmax" (softmax normalization)

Version: "ss" (simple span), "cs" (complex span), or "custom" (custom model)
→ activation functions in `bmmformula()` already implemented for "ss" and "cs"

id	setsize	setsize_lin	n_IIP	n_IOP	n_NPL	IIP	IOP	NPL
1	2	-3	1	2	0	23	0	0
10	2	-3	1	2	0	23	0	0
11	2	-3	1	2	0	23	0	0

(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmmodel: Choice rule

Memory Measurement Model

```
m3(resp_cats, num_options, choice_rule, version):
```

Different normalization methods: simple and softmax Luce's choice rule ("simple" or "softmax"):

Simple Luce's choice rule

Probability of response category i:

$$P(i) = \frac{A(i)}{\sum_{j=1}^{n} A(j)}$$

From all possible response categories j

- Absolute 0
- Estimation less stabile

Softmax Luce's choice rule

Probability of response category i:

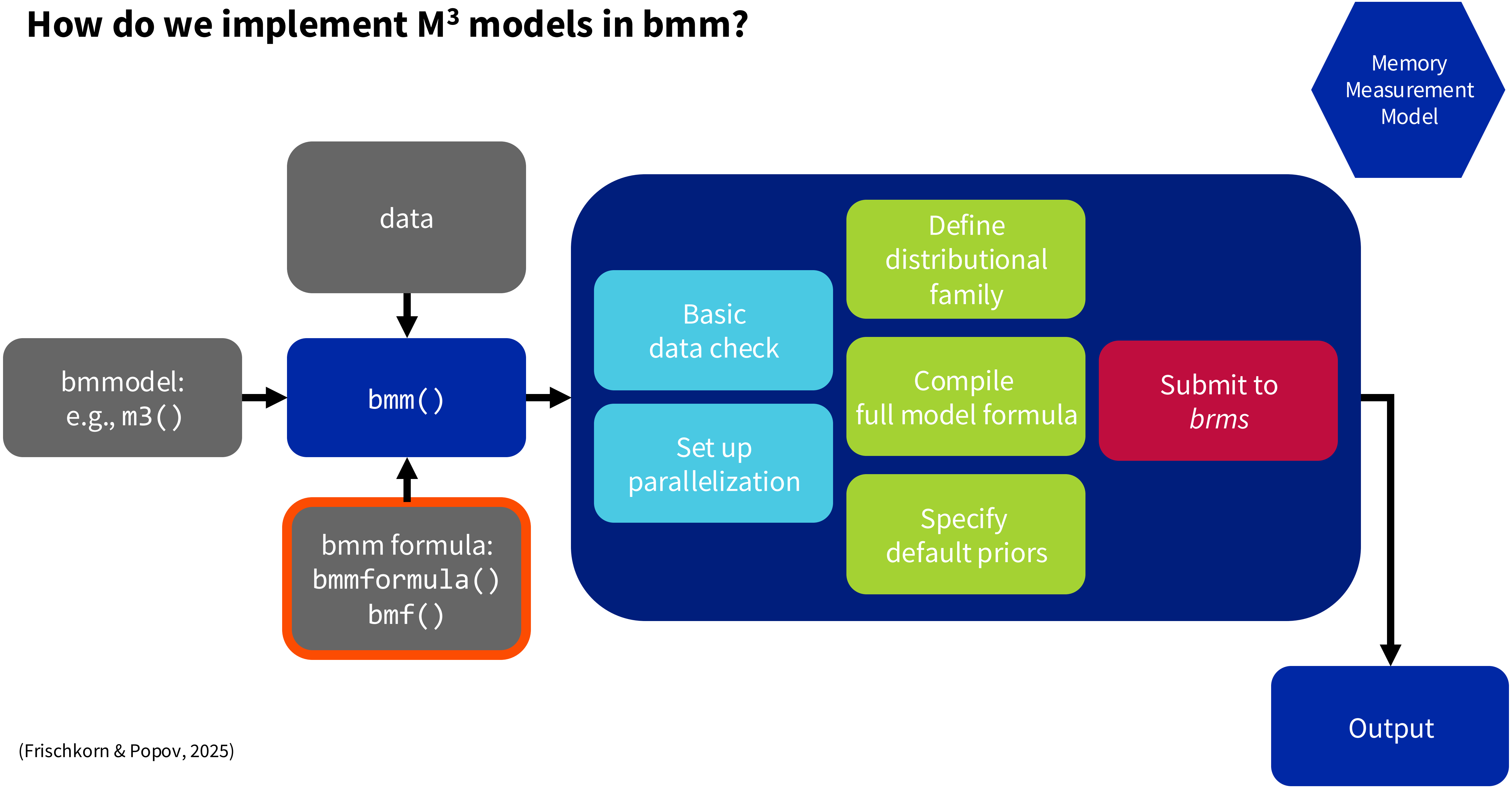
$$P(i) = \frac{e^{A(i)}}{\sum_{j=1}^{n} e^{A(j)}}$$

From all possible response categories j

- Relative 0
- Estimation more stabile

(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?



(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmm formula

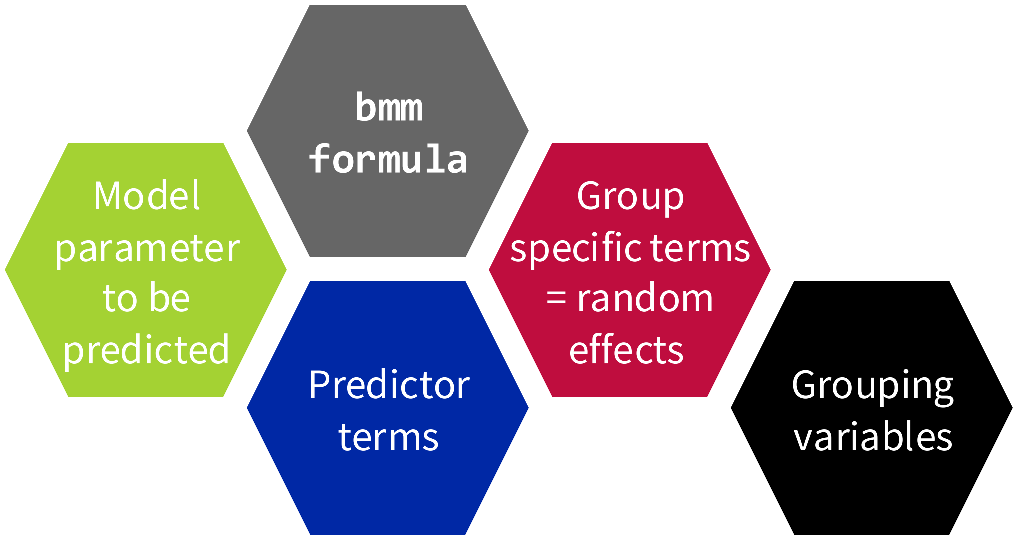
Memory
Measurement
Model

- Formulas predicting the different model parameters

- Be explicit and provide a formula for all model parameters

Important:
If you do not specify a formula for one of the model parameters only a fixed intercept will be estimated.

```
my_formula <- bmmformula(
    model_par1 ~ pterms + (gterms | group),
    model_par2 ~ pterms + (gterms | group)
)
```

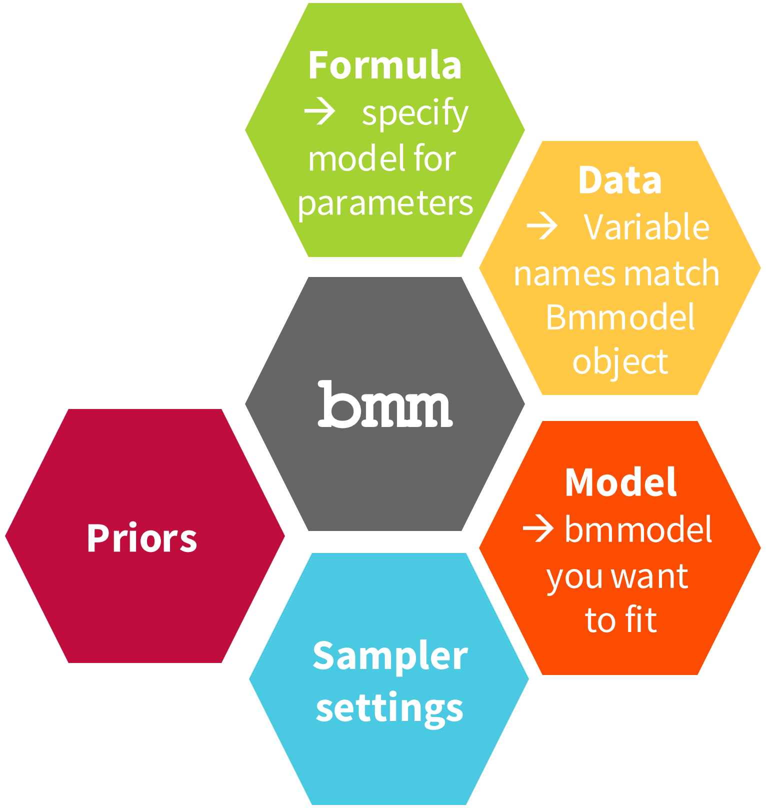


(Frischkorn & Popov, 2025)

How do we implement M³ models in bmm?

bmm() function

Memory
Measurement
Model




```
bmm(formula,
    data = myData,
    model,

    # optional arguments

    prior = myPriors,
    iter = 4000,
    warmup = 1000,
    chain = 4
)
```

Formula: bmm formula object that you created with `bmmformula()` or `bmf()`

Data: Data with all relevant variables

Model: bmmodel object that you created, e.g., with `m3()`

Prior: your priors that you define, e.g., `prior()`

Sampler setting: number of iterations, number of warm-ups, number of chains

(Frischkorn & Popov, 2025)

How do we assess M³ models in bmm?

Memory
Measurement
Model

- `summary()`
  show the model output

- `plot()`
  plots posterior distributions for every model estimate and the posterior samples per chain
  → check model convergence

- `pp_check()`
  posterior predictive plots to check model fit

- `conditional_effects()`
  plots marginal effects (conditional effects)

- `hypothesis()`
  performs hypothesis test → Evid.Ratio = BF_{01}

(Bürkner, 2017)

References

Memory Measurement Model

Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*, *80*(1), 1–28. https://doi.org/10.18637/jss.v080.i01

Frischkorn, G., & Popov, V. (2025). A tutorial for estimating mixture models for visual working memory tasks in brms: Introducing the Bayesian Measurement Modeling (bmm) package for R. *Behavior Research Methods, 57, 144.* https://doi.org/10.3758/s13428-025-02643-0

Oberauer, K., & Lewandowsky, S. (2019). Simple measurement models for complex working-memory tasks. *Psychological Review*, *126*(6), 880–932. https://doi.org/10.1037/rev0000159

How do we implement M³ models in bmm?

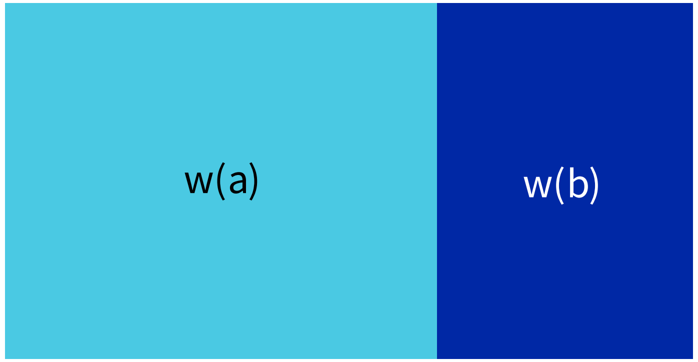
Bonus: Choice rule

Memory
Measurement
Model

- `m3(resp_cats, num_options, choice_rule, version):`
  - Different normalization methods: simple and softmax Luce's choice rule

Simple Luce's choice rule

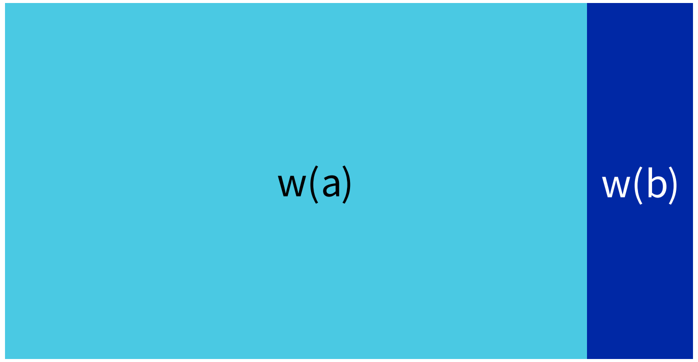
$$P(i) = \frac{A(i)}{\sum_{j=1}^{n} A(j)}$$



P (a) = 0.63

Softmax Luce's choice rule

$$P(i) = \frac{e^{A(i)}}{\sum_{j=1}^{n} e^{A(j)}}$$



P (a) = 0.88

Example:
response categories a and b

$w(a) = 5$
$e^{w(a)} = e^5 = 148$

$w(b) = 3$
$e^{w(b)} = e^3 = 20$

(Frischkorn & Popov, 2025)